performance of stocks in a centralized system. A nation's trade deficit fluctuates depending on reports from the distributed entities representing the various trade sectors. The world we live in is an excellent example of a naturally decentralized system, with its population determined by the global number of births and deaths. Take a few minutes to reflect on these examples.

Table 2.1 Examples of counters in diverse systems

System type	Counter example
Manual system	Turnstile for counting in an amusement park
Centralized system	Stock index
Distributed system	Nation's trade deficit
Decentralized system	World population

The counter is a simple but versatile use case that illustrates the smart contract development. You may be tempted to jump in and start coding, but you must resist this temptation and instead design the contract first. It's important to have the correct design before developing the code. Moreover, a design representation is independent of the smart contract language, thus providing you a blueprint for different implementations.

A smart contract is deployed on the blockchain by a transaction. It is permanently recorded on the blockchain, irreversible and unchangeable, and part of the chain of blocks, as stated in design principle 1.

> **DESIGN PRINCIPLE 1** Design before you code, develop, and deploy a smart contract on a test chain, and thoroughly test it before you